Practical Week -2

Task-1

Develop a C program that uses the system calls open(), read(), write(), and closeo to copy the contents of one file to 3 another. Explain how control transitions between user Space and kernel space during execution.

mohana@MOHANA: ~/task2.c

```
 GNU nano 8.7.1                task2.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd1, fd2;
    char buffer[100];
    int n;

    fd1 = open("File1.txt", O_RDONLY);

    if (fd1 < 0) {
        printf("Error opening File1.txt\n");
        return 1;
    }

    fd2 = open("File2.txt", O_CREAT | O_WRONLY | O_TRUNC, 0644);

    if (fd2 < 0) {
        printf("Error opening File2.txt\n");
        close(fd1);
        return 1;
    }

    n = read(fd1, buffer, sizeof(buffer));

    if (n < 0) {
        printf("Error reading file\n");
        close(fd1);
        close(fd2);
        return 1;
    }
 ^G Help          ^O Write Out     ^F Where Is
 ^K Cut           ^T Execute       ^X Exit
 ^R Read File     ^\ Replace       ^U Paste
 ^J Justify
```

mohana@MOHANA: ~/task2.c

```
 GNU nano 8.7.1                task2.c *
    if (n < 0) {
        printf("Error reading file\n");
        close(fd1);
        close(fd2);
        return 1;
    }

    write(fd2, buffer, n);

    close(fd1);
    close(fd2);

    printf("File copied successfully.\n");

    return 0;
}
 ^G Help          ^O Write Out     ^F Where Is
 ^K Cut           ^T Execute       ^X Exit
 ^R Read File     ^\ Replace       ^U Paste
 ^J Justify
```

mohana@MOHANA: ~

```
mohana@MOHANA:~$ echo "Hello, this is File1.">File1.txt
mohana@MOHANA:~$ cat File1.txt
Hello, this is File1.
mohana@MOHANA:~$ nano task2.c
mohana@MOHANA:~$ gcc task2.c -o task2
mohana@MOHANA:~$ ./task2
File copied successfully.
mohana@MOHANA:~$ cat File2.txt
Hello, this is File1.
mohana@MOHANA:~$ ls -l File.txt File2.txt
ls: cannot access 'File.txt': No such file or directory
File2.txt
mohana@MOHANA:~$ ls -l File1.txt File2.txt
-rw-r--r-- 1 mohana mohana 22 Aug 16 11:27 File1.txt
-rw-r--r-- 1 mohana mohana 22 Aug 16 11:33 File2.txt
```

Task-2

Use the strace utility to trace all system calls generated by the command cat sample.txt. Analyze the sequence of system 3 calls and identify the kernel services involved.

```
mohana@MOHANA: ~
mohana@MOHANA:~$ echo "Hello Operating System">sample.txt
mohana@MOHANA:~$ cat sample.txt
Hello Operating System
mohana@MOHANA:~$ strace -e trace=execve,openat,read,write,close,exit_group cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffc055e30c8 /* 25 vars */) = 0
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"..., 832) = 832
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"..., 832) = 832
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"..., 832) = 832
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0 \250\2\0\0\0\0\0"..., 832) = 832
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"..., 832) = 832
close(3)                                = 0
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbd"..., 1024) = 442
```

```
mohana@MOHANA: ~
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"..., 832) = 832
close(3)                                = 0
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbd"..., 1024) = 442
close(3)                                = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
read(3, "none /usr/lib/modules/6.18.33.2-"..., 1024) = 1024
read(3, "e 0 0\ndevpts /dev/pts devpts rw,"..., 1024) = 1024
read(3, ",access=client,msize=65536,trans"..., 1024) = 1024
read(3, "700,noswap 0 0\ntmpfs /run/user/1"..., 1024) = 248
read(3, "", 1024)                       = 0
close(3)                                = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
read(3, "55c56b713000-55c56b805000 r--p 0"..., 1024) = 1024
read(3, "0 08:30 13628                     "..., 1024) = 1024
read(3, "00 08:30 13631                     "..., 1024) = 1024
read(3, "                 /usr/lib/x86_64-lin"..., 1024) = 1024
read(3, "0 13625                  /us"..., 1024) = 566
close(3)                                = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
read(3, "# Common strings shared across a"..., 4080) = 4080
read(3, "", 32)                         = 0
close(3)                                = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
Hello Operating System
close(5)                                = 0
close(4)                                = 0
close(3)                                = 0
exit_group(0)                           = ?
+++ exited with 0 +++
mohana@MOHANA:~$
```